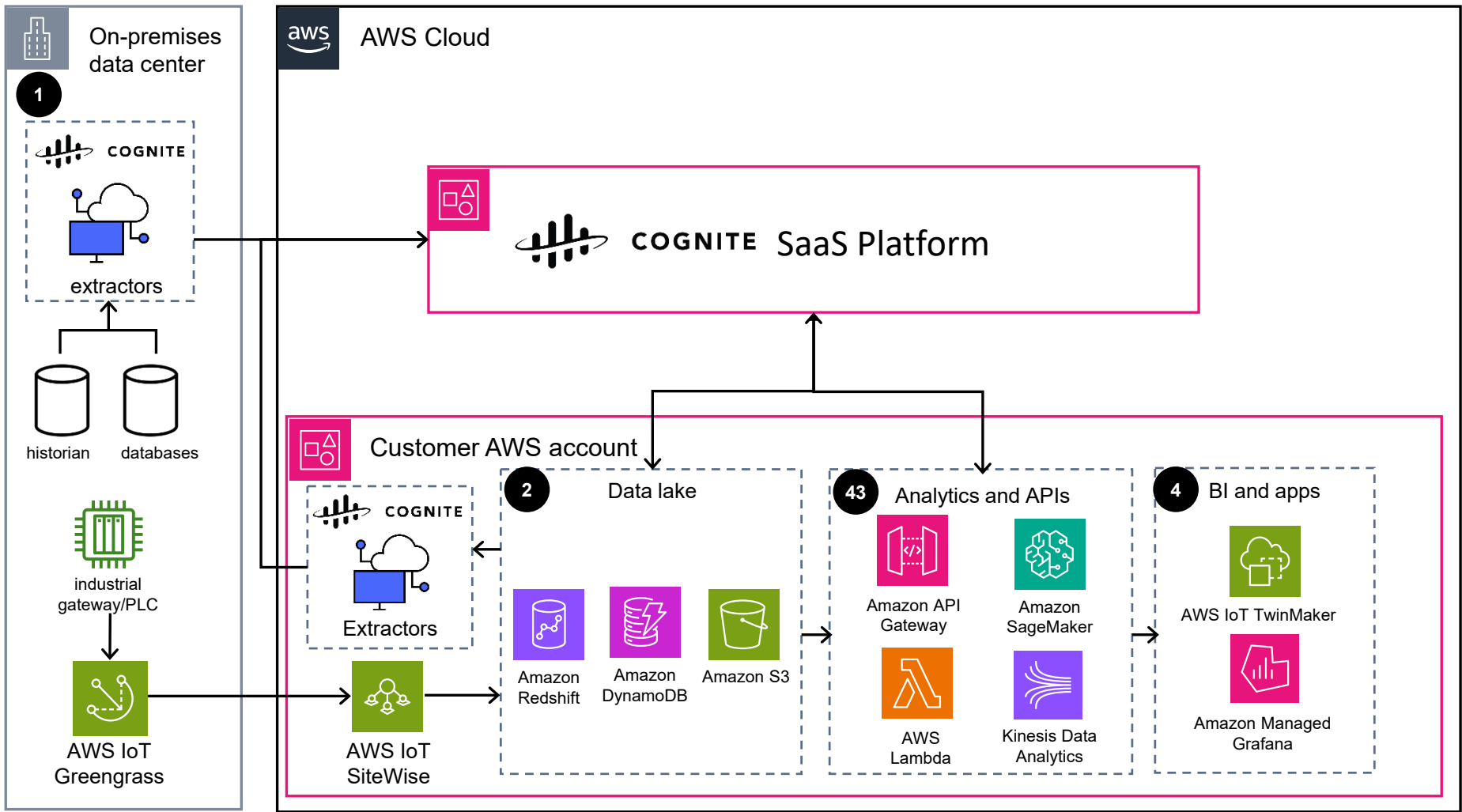


Industrial DataOps on AWS using Cognite Data Fusion

This reference architecture enables you to harness the power of AWS Cloud using [Cognite Data Fusion](#) (CDF), an industrial DataOps platform. CDF provides data contextualization that helps customers leverage AWS services to build differentiated capabilities, enabling data-driven decisions for industrial operations. It collects diverse industrial data sources such as OT, IT, and ET data, contextualizes the information, and develops industrial knowledge graphs and applications. These tools help maximize production efficiency, ensure safe and sustainable operations, and enable high-quality, AI-powered business decisions.



- 1** Purpose built extractors ingest industrial data sources such as IT, OT and Engineering Data or use **AWS IoT Greengrass**, an open source IoT edge runtime to bring data from Edge Gateways or PLCs to AWS cloud.
- 2** Data from industrial historians and other industrial devices are ingested through **AWS IoT SiteWise**. Cognite Native Extractors ingest pre-aggregated industrial data from customer's data lake based in either **Amazon Simple Storage Service (Amazon S3)**, **Amazon Redshift**, and **Amazon DynamoDB**.
- 3** Customers use **AWS Lambda** and **Amazon API Gateway** together to ingest data from CDF, transform data using **AWS Glue**, and write data back into CDF using the Cognite SDK. Additional analytics and machine learning capabilities are provided by **Amazon Kinesis Data Analytics** and **Amazon SageMaker**, respectively.
- 3** End users create custom applications depending on data type using **AWS IoT TwinMaker** and **Amazon Managed Grafana**.